Javed N

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PROFESSIONAL SUMMARY

Senior Data Science & AI/ML Lead with 6 years of experience building end-to-end machine learning systems, Generative AI solutions, and production-grade analytics products across multiple domains. Proficient in Python, PySpark, cloud-native ML (AWS, Azure, GCP), deep learning, NLP, LLMs, RAG architectures, and MLOps pipelines. Proven ability to translate ambiguous business problems into scalable AI products - from data engineering and feature design to model development, deployment, monitoring, and continuous improvement. Strong background in supervised learning, time-series forecasting, optimization, and enterprise ML architecture. Experience in partnering with cross-functional engineering, product, and business teams to deliver measurable impact, automate decision workflows, and ship reliable, high-impact AI systems at scale. IIT Bombay Dual Degree + IIM Indore MBA with strong quantitative and business fundamentals with additional expertise in enterprise-level credit & market risk modelling, stochastic analysis, GenAI-driven automation pipelines, and AI/data governance frameworks.

EDUCATIONAL QUALIFICATIONS

MBA	Finance & Analytics, IIM Indore	3.17/4.33	2018-20
B.Tech + M.Tech + Minor	Aerospace Eng. (Major) & Electrical Eng. (Minor), IIT Bombay	8.52/10 (B.Tech) | 9/10 (M.Tech)	2010-15

Certifications	AWS Certified Machine Learning Engineer Associate · KPMG Six sigma · NISM Research Analyst · Microsoft Office Specialist · Decision Intelligence (LinkedIn) · FRM (Financial Risk Manager)
Programming & Data Engineering	Python · PySpark · SQL · R · Java · MATLAB · Data Pipelines (Azure Data Factory, AWS Glue) · ETL/ELT · Flask APIs · Git · CI/CD · Data Wrangling · Feature Engineering · Distributed Computing
Machine Learning	Supervised Learning (Classification, Regression) · Time-Series Forecasting (ARIMA, Prophet, GBM/O-U processes) · Feature Selection & Model Explainability (SHAP, LIME) · Ensemble Methods (XGBoost, LightGBM, CatBoost) · Optimization · A/B Testing · Statistical Inference · Experimentation at Scale
Deep Learning & NLP	PyTorch · TensorFlow · Keras · HuggingFace Transformers · Sequence Models (LSTM, GRU) · BERT-style Models · Embeddings · Text Classification · Information Extraction · Named Entity Recognition (NER) · Semantic Search
Cloud & MLOps	AWS (SageMaker, EC2, Lambda, S3) · Azure (ADF) · GCP (Vertex AI, BigQuery) · Model Deployment · Monitoring & Drift Detection · Containerization (Docker) · Cloud-native ML Services
Generative AI & LLM Engineering	Prompt Engineering (CoT, Few-Shot, Zero-Shot) · LLM Fine-Tuning (LoRA/PEFT) · RAG Architecture (FAISS, Chroma, LangChain, Vector DBs) · LLM-driven AI Agents · Multi-step Reasoning Pipelines · OpenAI API
Visualization/ Front-end	PowerBI · Tableau · ThoughtSpot · Streamlit (Internal UIs)
Credit Risk & Financial Analytics	PD/LGD Modeling · Survival Analysis (Time-to-Default) · Credit Scorecards · Portfolio Risk & NPA Forecasting · Stress Testing · Exposure & Cashflow Modeling · Regulatory & Risk Reporting (Basel/IFRS concepts)

ACADEMIC & PROFESSIONAL ACHIEVEMENTS

▪ Leadership Excellence Award for **Exceptional Leadership** across Tech domain consisting of 800 people in RBL Bank	2022
▪ Ranked among **top 5% students** in both IIT Bombay and IIM Indore, Awarded branch change in IIT Bombay given to Top 2%	2020

PROFESSIONAL EXPERIENCE — 70 months

Marsh McLennan — **Senior Data Scientist Lead – Risk Consulting/ AI Engineering** — *July'24 – Present*

- **AI-driven Valuation Data Extraction platform**
  Built a production-grade RAG-based document intelligence platform using OpenAI, LangChain, FAISS/Chroma for vector search, enabling automated extraction of structured insights from 100+ complex unstructured valuation documents. Engineered embeddings, chunking pipelines, and multi-stage prompt orchestration to deliver accurate context retrieval, entity extraction, and summarization. Deployed the system using AWS (EC2, S3) + Flask API + Streamlit UI**, reducing manual analyst effort by 3+ hours per document and enabling scalable document-processing workflows.**
- **Climate Analysis and Risk Advisory Tool**
  Architected and deployed a cloud-native **AI driven climate analytics platform** integrating geospatial datasets, and domain rules to quantify various asset-level risk factors (e.g., physical, environmental, operational). Enhanced the value proposition using Agentic AI Framework by building Hierarchical Agent framework which uses conversational AI Agents built on domain specific documents as context along with Vision AI Modules. Built Machine Learning components for hazard scoring, anomaly detection, and simulation-based risk mapping using Python, GeoPandas, and ML frameworks. Deployed as a containerized application using **Docker on AWS**, enabling API-based consumption and integration with internal analytics tools. Product delivered **>$1 million in consulting revenue** within India and Middle East Region, demonstrating strong ability to build **market-ready AI solutions** end-to-end.
- **Financial Impact Forecasting & Scenario Modelling tool for Enterprise Risk and Valuation**
  Developed an AI-powered **stochastic forecasting engine** (GBM, OU processes) integrated with LLM RAG pipeline for automated extraction of financial data. **Improved forecasting accuracy by around 20-25%**, accelerated capital planning, and linked multi-risk modules to quantify P&L and cashflow impact across scenarios. The tool was deployed as an **internal decision-support product** used by risk advisory and strategy teams to rapidly evaluate the financial resilience of client portfolios. Collaborated with cross-functional stakeholders to define product requirements, build scenario templates, integrate regulatory stress assumptions, and design an intuitive interface for model outputs. The solution automated previously manual forecasting workflows, **reduced turnaround time for financial analysis by over 50%,** and enabled seamless comparison of multi-scenario P&L and cashflow outcomes. This product is now used as a repeatable, scalable framework for enterprise-risk discussion with large global clients with **>$2 million revenue generated.**

RBL Bank	Data Scientist / Analytics Manager	*Oct'20 – July'24*

- **Home Loan Credit Risk Scorecard (PD Model + Survival Analysis)**
  Led the development of an end-to-end predictive modelling platform to assess customer credit behaviour and default likelihood using a combination of machine learning, survival analysis, and credit bureau datasets. Engineered a comprehensive feature pipeline by integrating CIBIL bureau data, repayment patterns, loan behaviour, utilization ratios, and inquiry activity across the customer lifecycle. Performed extensive exploratory analysis, feature engineering, model training, and validation. The final system combined classification and time-to-default modelling to generate more granular risk signals and was deployed using Vertex AI with automated drift monitoring and CI/CD workflows. This solution improved **risk segmentation accuracy by ~20%,** contributing to a **10% reduction in observed default rates** and materially enhanced decisioning quality across origination and portfolio review functions.
- **Credit Cards Acquisition Propensity Modeling and Cross-Sell Recommender System**
  Designed and implemented Machine Learning based acquisition models to identify high-propensity customers for credit card origination. By leveraging behavioural patterns, transaction data, and historical campaign performance, built predictive models that improved targeting efficiency. Additionally, developed a recommender system using Apriori algorithm, content and collaborative filtering for up-sell/cross-selling products to customers leading to **~15% cost reduction per campaign** and **~20 increase in conversion**.
- **Portfolio NPA, Cashflow and Provision Forecasting Framework**
  Developed a robust forecasting framework to project NPA evolution, portfolio-level cashflows, and expected credit losses for multiple asset classes. By analyzing historical delinquency, recovery, and cure patterns, established a statistical baseline and then introduced advanced time-series models such as ARIMA and Prophet to capture complex temporal dynamics. The solution included scenario overlays - benign, moderate stress, and severe stress to support provisioning and risk appetite decisions. **The final model delivered ~15–20% improvement in forecasting accuracy** and enabled the risk team to better estimate monthly slippages, provisions, recoveries, and capital adequacy requirements, strengthening the bank's overall portfolio monitoring capability.
- **Portfolio MIS, Dashboards, and Risk Visualization**
  Led a team to create portfolio MIS dashboards using PowerBI and ThoughtSpot to track NPA trends, provision utilization, recovery performance, and delinquency progression across time. Automated reporting workflows end-to-end, significantly reducing the manual effort required for senior management updates.
- **Real time CIBIL Trigger and Early Warning System**
  Built an automated real-time bureau-trigger pipeline integrated with the CIBIL Watch API. This system captured bureau inquiries, delinquency alerts, utilization changes, and credit line movements as soon as they occurred. I developed ETL workflows using SQL and Azure Data Factory to route risk triggers into CRM and contact centre systems for immediate outreach. This real-time signal-based monitoring allowed the bank to **proactively intervene on high-risk borrowers and contributed to loan disbursal worth ~50 Cr.**
- **Regulatory Reporting and Data Governance for Credit Risk**
  Strengthened **regulatory and governance reporting** by streamlining key SQL and Python-based data flows, improving data lineage, and creating reusable ETL frameworks for critical reports such as NPA statements, provision summaries, delinquency buckets, and capital adequacy metrics. Worked closely with the risk governance team to ensure alignment with Basel and IFRS concepts, validation rules, and audit controls. These enhancements ensured higher data quality and consistency across all regulatory submissions.

Purplle.com	Software Engineer	2015-16

Developed data-driven in app features in app to support financial and operational decision-making for e-commerce transactions.

	Professional Exploration	2016-17

Engaged in comprehensive exploration of the startup ecosystem, derived valuable business insights, and published a research paper.

INTERNSHIPS

Servify, Mumbai	Product Management	*Apr '19 – Jun '19*

Authored a detailed Product Lifecycle Management (PLM) note for chatbot, QMS to improve product quality and customer experience.

GITHUB PROJECT

Algorithmic Trading & Machine Learning Based Forecasting Toolkit - GitHub: https://github.com/javed18/Quantitative-Stock-Trading

Developed a comprehensive quantitative trading and forecasting toolkit integrating machine learning, statistical modelling, and financial analytics. The system enables users to load market data, engineer technical features, generate portfolio level strategies and back-test strategies to evaluate performance.

KEY CAPABILITIES

End-to-end ownership of AI/ML products and credit/market risk models (problem framing → modelling → deployment → monitoring); ability to architect scalable cloud-native ML systems (AWS, Azure, GCP) with strong foundations in supervised learning, NLP/LLMs, time-series, and RAG-based automation; proven experience translating business requirements into production-grade AI solutions; additional expertise in enterprise credit risk modelling, cashflow modelling, Financial modelling, climate risk modelling and deployment.

PUBLISHED RESEARCH PAPERS & CASE STUDIES

Simulation of Swarm Robots mission control – IFAC (2016)	**'Yes Bank - The Cost of Saying Yes' – AIMA Case Study (2022)**

LEADERSHIP & ACTIVITIES

Senior Member – UTSAHA Projects Vertical, IIM Indore. Coordinator – Mood Indigo & Techfest, IIT Bombay.

Gold medallist, Chess (IIM Indore inter-college fests).

LinkedIn Creator with **1 million+ content impressions**; financial markets; chess; reading; painting.

TECHNICAL STRENGTHS

Experienced in cloud-native ML pipelines (Python, PySpark, Docker, CI/CD), experiment tracking, model monitoring, and drift detection; skilled in API-based ML deployment (Flask/FastAPI), serverless compute, and containerized workloads across AWS/GCP/Azure; proficient with LLM/RAG systems, vector databases, and integrating classical ML with GenAI; capable of designing automated ETL workflows, feature stores, and high-quality data pipelines.